

Rare Earth Element (REE) Geochemistry: Chondrite Normalization, The Europium Anomaly & Plagioclase Partitioning

Comprehensive Geochemistry Guide · Authoritative Reference: John D. Winter, Principles of Igneous and Metamorphic Petrology (2nd Ed., Ch. 9)

THEORETICAL FOUNDATION: REE SYSTEMATICS & ANOMALY FORMATION

Lanthanide Contraction, Oddo-Harkins Smoothing, Europium Valence Multiplicity & The Sm-Gd Baseline

1. The Rare Earth Elements (Lanthanides) & Chondrite Normalization

The Rare Earth Elements (REE) comprise the 15 elements with atomic numbers $Z = 57$ (Lanthanum, La) through $Z = 71$ (Lutetium, Lu). Promethium (Pm , $Z = 61$) has no stable isotopes and is absent in nature. Across this series, electrons fill the inner $4f$ subshell while the outer valence shell remains constant:

The Lanthanide Contraction: As atomic number increases from La ($Z=57$) to Lu ($Z=71$), increasing nuclear charge pulls the electron cloud closer, causing ionic radii to shrink smoothly from $1.16 \text{ \AA}$ (La^{3+}) down to $0.977 \text{ \AA}$ (Lu^{3+}). This creates a predictable gradient in chemical behavior: Light REE (LREE: La to Sm) are larger and more incompatible, while Heavy REE (HREE: Gd to Lu) are smaller and more compatible in minerals like garnet.

Because nucleosynthetic processes favor even-atomic-number nuclei (the **Oddo-Harkins Effect**), raw REE abundances display an irregular "sawtooth" pattern. Geochemists normalize rock abundances against primitive carbonaceous chondrite meteorites (CI chondrites) to produce a smooth curve:

CHONDRITE NORMALIZATION FORMULATION

$$\left[\frac{\text{REE}_i}{\text{sample}} \right]_N = \frac{[\text{REE}_i]_{\text{sample}}}{[\text{REE}_i]_{\text{chondrite}}}$$

Plotting $\left[\frac{\text{REE}_i}{\text{sample}} \right]_N$ on a logarithmic y-axis against atomic number (Z) generates a smooth, diagnostic REE pattern. Any deviation from a smooth curve reflects an anomalous geochemical process affecting that specific element.

2. Sequential Order of REEs: Locating Europium Between Sm and Gd

In standard REE diagrams, elements are plotted in strict order of increasing atomic number along the horizontal axis:

Group	Element	Symbol	Atomic Number (Z)	Common Valence	Ionic Radius (Å)
LREE	Lanthanum	La	57	+3	1.160
LREE	Cerium	Ce	58	+3, +4 (anomalous)	1.143
LREE	Praseodymium	Pr	59	+3	1.126
LREE	Neodymium	Nd	60	+3	1.109
—	Promethium	Pm	61	Radioactive / Omitted	—
LREE / MREE	Samarium	Sm	62	+3	1.079
Anomaly Site	EUROPIUM ★ Subject of Anomaly	Eu	63	+2, +3 (multivalent)	Eu^{2+} : 1.250 Eu^{3+} : 1.066
HREE / MREE	Gadolinium	Gd	64	+3	1.053
HREE	Terbium	Tb	65	+3	1.040
HREE	Dysprosium	Dy	66	+3	1.027

Structural Sequence: Notice that **Europium (Z = 63)** is situated precisely between **Samarium (Z = 62)** and **Gadolinium (Z = 64)**. Therefore, on every standardized REE plot, the Europium anomaly appears as a vertical spike or trough situated directly in the interval between Sm and Gd .

3. Textbook Evidence: The Europium Anomaly on REE Plots

Figure 4 from Winter illustrates chondrite-normalized REE patterns for basaltic and andesitic suites, displaying positive and negative Europium anomalies:

TABLE 5 Spreadsheet to Calculate an REE Diagram for th					
	A	B	C	D	E
1	Rare Earth Element Diagram				
2	Element	Atom #	chondr	sample	samp/ch
3	La	57	0.237	1.75	7.40
4	Ce	58	0.612	4.86	7.94
5	Nd	60	0.467	3.37	7.22
6	Sm	62	0.153	0.98	6.43
7	Eu	63	0.058	0.15	2.62
8	Dy	66	0.254	1.00	3.94
9	Er	68	0.1655	0.44	2.65
10	Yb	70	0.17	0.37	2.17
11	Lu	71	0.0254	0.05	2.00
12					
13			in ppm	in ppm	
14					
15			Distribution Coefficients		
16		Olivine	Opx	Cpx	Garnet
17	La	0.007	0.03	0.056	0.00
18	Ce	0.006	0.02	0.092	0.00
19	Nd	0.006	0.03	0.230	0.02
20	Sm	0.007	0.05	0.445	0.10
21	Eu	0.007	0.05	0.474	0.24
22	Dy	0.013	0.15	0.582	3.17
23	Er	0.026	0.23	0.583	6.56
24	Yb	0.049	0.34	0.542	11.5
25	Lu	0.045	0.42	0.506	11.9

Figure 1 (Winter Fig. 4, p. 173): Chondrite-normalized REE diagram. Elements are arranged in order of atomic number from La to Lu . Note the dramatic vertical excursion at **Europium (Eu)**, situated between **Sm** and **Gd** . A sharp dip represents a *negative Eu anomaly* ($\text{Eu}/\text{Eu}^* < 1$), while an upward spike represents a *positive Eu anomaly* ($\text{Eu}/\text{Eu}^* > 1$).

time. As I have suggested, this is a common problem for most of our models. Using them, we might be able to identify

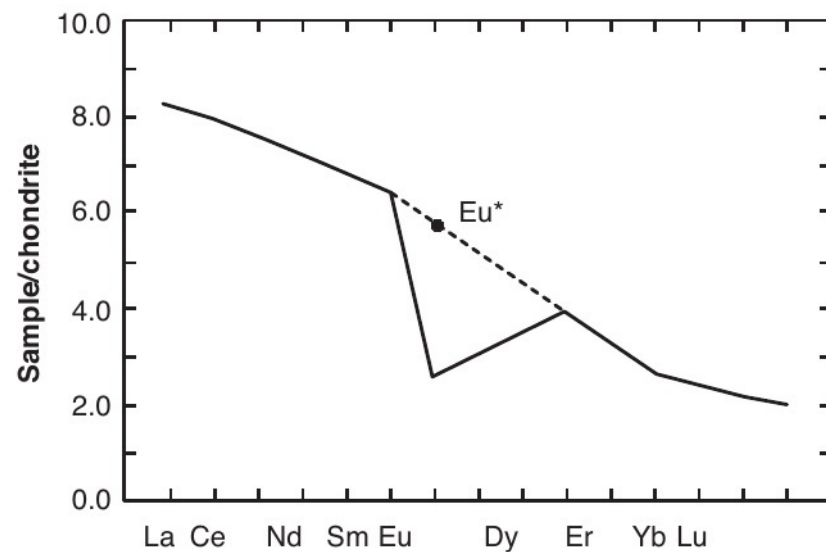


FIGURE 5 REE diagram for 10% batch melting of a hypothetical lherzolite with 20% plagioclase. Because Eu^{2+} is retained in the plagioclase of the source, the extracted melt is depleted in Eu, resulting in a pronounced negative europium anomaly.

Figure 2 (Winter Fig. 5, p. 174): Mineral/melt partition coefficients for REEs. For all trivalent REEs (Sm^{3+} , Gd^{3+} , etc.), $D_{\text{plagioclase}}$ is low ($0.05\text{--}0.1$). However, at Europium, $D_{\text{plagioclase}}$ jumps to >1.5 because divalent Eu^{2+} substitutes freely for Sr^{2+} and Ca^{2+} in the plagioclase crystal lattice.

4. The Redox Mechanism: Why Europium Behaves Uniquely

Under terrestrial magmatic oxygen fugacities (typically around the QFM buffer), Europium exists in a multivalent equilibrium:

THE EUROPIUM REDOX EQUILIBRIUM & ANOMALY METRIC

$\text{Eu}^{3+} + e^- \rightleftharpoons \text{Eu}^{2+}$

- In the trivalent state (Eu^{3+}), its radius is $1.066 \text{ \AA}$, nearly identical to Sm^{3+} and Gd^{3+} .
- In the divalent state (Eu^{2+}), its radius swells to $1.250 \text{ \AA}$, matching Sr^{2+} ($1.18 \text{ \AA}$) and Ca^{2+} ($1.00 \text{ \AA}$).

Because plagioclase accepts Ca^{2+} and Sr^{2+} , it preferentially absorbs Eu^{2+} ($D_{\text{plag}}^{\text{Eu}} \gg 1$), while rejecting all other REEs ($D_{\text{plag}}^{\text{REE}^{3+}} \ll 1$).

Mathematical Calculation of the Anomaly (Eu/Eu^*):

$\text{Eu}^* = \sqrt{\text{Sm}_N \times \text{Gd}_N} \quad \text{or} \quad \text{Eu}^* = \frac{\text{Sm}_N + \text{Gd}_N}{2}$

$\frac{\text{Eu}}{\text{Eu}^*} = \frac{\text{Eu}_N}{\sqrt{\text{Sm}_N \times \text{Gd}_N}}$

- $\text{Eu}/\text{Eu}^* < 1.0$: **Negative anomaly** (plagioclase removed by fractional crystallization).
- $\text{Eu}/\text{Eu}^* > 1.0$: **Positive anomaly** (plagioclase accumulated, e.g. anorthosite).

"The magnitude of the Eu anomaly is commonly expressed as Eu/Eu^ , where Eu^* is the hypothetical value of Eu if no Eu^{2+} was present, determined by interpolating between the chondrite-normalized values of Sm and Gd ... A negative europium anomaly is a good indicator that the liquid was at one time in equilibrium with now-absent plagioclase."*

— John D. Winter, *Principles of Igneous and Metamorphic Petrology*, Chapter 9, p. 175.



EXAM QUESTION & RIGOROUS OPTION-BY-OPTION DISSECTION

DGMS 2019 Examination · Official Question Statement, Verified Answer & Detailed Distracter Analysis

DGMS 2019 · QUESTION 21

In a rare earth element (REE) diagram, europium anomaly is seen between which one of the following REE pairs?

- (a) La-Nd
- (b) Nd-Sm
- (c) Sm-Gd
- (d) Gd-Dy



Correct Option: (c) Sm-Gd

Verified via John D. Winter, Principles of Igneous & Metamorphic Petrology (Ch. 9, pp. 173–175)

INCORRECT Option (a): La-Nd

Detailed Refutation: Lanthanum (La , $Z=57$) and Neodymium (Nd , $Z=60$) are both Light Rare Earth Elements. The elements intervening between them are Cerium (Ce , $Z=58$) and Praseodymium (Pr , $Z=59$). While a *Cerium anomaly* (Ce^{Ce^*}) can occur in marine environments between La and Pr due to oxidation of Ce^{3+} to insoluble Ce^{4+} , Europium ($Z=63$) does not lie anywhere near this interval.

INCORRECT Option (b): Nd-Sm

Detailed Refutation: Neodymium (Nd , $Z=60$) and Samarium (Sm , $Z=62$) flank the synthetic/radioactive element Promethium (Pm , $Z=61$). Promethium does not occur in natural geological samples. No Europium anomaly can exist here, as Europium has atomic number $Z=63$, which lies beyond Samarium.

CORRECT ANSWER **Option (c): Sm-Gd**

Detailed Validation: Europium (Eu , $Z = 63$) is positioned directly between Samarium (Sm , $Z = 62$) and Gadolinium (Gd , $Z = 64$) in the periodic table and along the x-axis of every standard chondrite-normalized REE diagram. Because Sm and Gd exist strictly as trivalent cations (Sm^{3+} , Gd^{3+}), they establish the smooth, undisturbed background baseline. Europium's multivalent behavior (Eu^{2+} entering plagioclase) causes its normalized concentration to deviate sharply above or below this baseline, creating the diagnostic **Europium anomaly precisely between Sm and Gd** . The baseline value Eu^* is mathematically calculated as the geometric mean $\sqrt{\text{Sm}_N \times \text{Gd}_N}$.

INCORRECT **Option (d): Gd-Dy**

Detailed Refutation: Gadolinium (Gd , $Z=64$) and Dysprosium (Dy , $Z=66$) are Heavy Rare Earth Elements that flank Terbium (Tb , $Z=65$). All three elements (Gd , Tb , Dy) are strictly trivalent and define the smooth downward or flat trajectory of the HREE profile, completely past Europium.



SYNTHESIS & HIGH-YIELD PRACTICE QUESTIONS

Curated Problems Covering REE Modeling, Garnet Signatures, Cerium Anomalies & Lunar Geochemistry

High-Yield Exam Practice Questions

Practice Q1 (Numerical / Anomaly Calculation): Quantifying Eu/Eu*

A rhyolitic lava displays chondrite-normalized abundances of $\text{Sm}_N = 36.0$, $\text{Eu}_N = 6.0$, and $\text{Gd}_N = 16.0$. Calculate the Europium anomaly (Eu/Eu^*) and state whether it is positive or negative.

Solution & Rationale:

Interpolated baseline: $\text{Eu}^* = \sqrt{\text{Sm}_N \times \text{Gd}_N} = \sqrt{36.0 \times 16.0} = \sqrt{576} = 24.0$.

$\frac{\text{Eu}}{\text{Eu}^*} = \frac{\text{Eu}_N}{\text{Eu}^*} = \frac{6.0}{24.0} = \mathbf{0.25}$.

Because $\text{Eu}/\text{Eu}^* = 0.25 < 1.0$, the rock displays a **severe negative Europium anomaly**, indicating extensive fractional crystallization and removal of plagioclase feldspar.

Practice Q2 (Planetary Petrology): The Lunar Magma Ocean

The lunar highlands (anorthosite crust) display a huge positive Europium anomaly, while lunar mare basalts display a pronounced negative Europium anomaly. How does this prove the existence of the primordial Lunar Magma Ocean?

Answer: During crystallization of the global Lunar Magma Ocean, low-density plagioclase floated to form the buoyant lunar anorthositic crust, scavenging Eu^{2+} and creating a massive **positive Eu anomaly**. The complementary residual liquid, depleted in Eu, sank into the lunar mantle; subsequent remelting of this residue formed mare basalts carrying the complementary **negative Eu anomaly**.

Practice Q3 (Marine Geochemistry): The Cerium Anomaly

Which other Rare Earth Element produces a prominent anomaly in modern seawater and pelagic sediments, what is its redox state, and which REE pair flanks it?

Answer: Cerium (Ce). Under oxidizing surface seawater conditions, soluble Ce^{3+} oxidizes to insoluble Ce^{4+} , precipitating into ferromanganese nodules. This leaves seawater with a strong **negative Ce anomaly** between La and Pr ($\text{Ce}^* = \sqrt{\text{La}_N \times \text{Pr}_N}$).

Practice Q4 (Garnet Signature): Slope of the REE Profile

A basalt has an extremely steep chondrite-normalized REE pattern with a very high $\frac{\text{La}}{\text{Yb}}_N$ ratio (>20) and heavy depletion in HREE (Yb , Lu). What residual mineral in the mantle source is diagnostic of this signature?

Answer: Garnet. Garnet has an extraordinarily high partition coefficient for heavy REEs ($D_{\text{garnet}}^{\text{Lu}} \approx 12$), locking them into the solid mantle residue. This proves that melting occurred deep within the garnet peridotite stability field ($>80 \text{ km}$ depth).

Practice Q5 (Redox Sensitivity): Eu Valence in the Upper Mantle vs. Terrestrial Atmosphere

Why does Europium exhibit anomalous divalent behavior in basaltic magma chambers, but not in low-temperature oxidizing surface waters?

Answer: Divalent Eu^{2+} is stable only under reducing conditions (low oxygen fugacity $f_{\text{O}_2} \leq \text{QFM}$ buffer), typical of deep magmatic chambers. In oxygenated surface waters, all Europium oxidizes to trivalent Eu^{3+} , behaving identically to its neighbors and producing no anomaly.

Practice Q6 (Plagioclase Accumulates): Anorthosites

An igneous cumulate rock composed of $>90\%$ bytownite plagioclase displays a chondrite-normalized REE profile. What characteristic feature will be observed at $Z = 63$?

Answer: A prominent **positive Europium anomaly** ($\frac{\text{Eu}}{\text{Eu}^*} \gg 1.0$), because the accumulated plagioclase crystals concentrated Eu^{2+} from the parental liquid.

Practice Q7 (Oddo-Harkins Rule): Nucleosynthetic Cause

What fundamental nuclear property explains why natural abundances of Ce , Nd , Sm , Gd are systematically higher than those of La , Pr , Eu , Tb in raw rock analyses?

Answer: The Oddo-Harkins Rule. Atomic nuclei with even atomic numbers ($Z = 58, 60, 62, 64$) have paired protons, providing greater nuclear binding energy and stability during stellar nucleosynthesis than nuclei with odd atomic numbers ($Z = 57, 59, 63, 65$).

Practice Q8 (Crustal Composition): Post-Archean Australian Shale (PAAS)

Why does the average upper continental crust (and PAAS) display a pervasive negative Europium anomaly ($\frac{\text{Eu}}{\text{Eu}^*} \approx 0.65$)?

Answer: The upper continental crust is dominated by differentiated granitic rocks and their sedimentary erosional products. These granites formed by partial melting or fractional crystallization that left plagioclase-rich residues in the lower crust, permanently depriving the upper crust of Europium.

Comprehensive Rare Earth Element Classification Matrix

Subdivision	Elements & Atomic Numbers	Ionic Radius (\$3+ \$)	Mantle Compatibility	Diagnostic Mineral Reservoir
Light REE (LREE)	La (57) to Nd (60)	Largest (1.16–1.11 Å)	Highly Incompatible	Enriched in continental crust & alkaline melts; depleted in N-MORB.
Middle REE (MREE)	Sm (62) to Ho (67)	Intermediate (1.08–1.02 Å)	Moderately Incompatible	Concentrated by amphibole (hornblende). Flanks the Eu anomaly (Sm-Eu-Gd).
Heavy REE (HREE)	Er (68) to Lu (71)	Smallest (1.00–0.98 Å)	Compatible in Garnet	Strongly retained by Garnet in deep mantle peridotite and eclogite.